MID-TERM EXAMINATION
B.Tech 1st semester (For students admitted in 2024)
(October,2024)
OFF LINE mode

Subject Code: BAS 101	Subject: Applied Mathematics
Time : 1 ½Hours	Maximum Marks : 30
Note : Q1 is compulsory. Calculator is not allowed.	

Q1		(2.5*4)
	(a) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$ by reducing it to echelon form.	CO1
	(b) Show that the eigen values of a Skew-Hermitian matrix are either zero or purely imaginary.	CO1
	(c) Find the n^{th} derivative of $e^{2x}\cos^2 x \sin x$.	CO2
	(d) Discuss the continuity of the function $f(x,y) = \begin{cases} x^3y/(x^4+y^4), & (x,y) = (0,0) \\ 0 & (x,y) \neq (0,0) \end{cases}$ at $(0,0)$.	CO2
Q2	**(Attempt any Two Parts) UNIT-1 (CO1)**	(5,5)
	(a) Using Gauss-Jordan method, find the inverse of the matrix $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$.	CO1
	(b) Solve the system of linear equations $3x_1 - 6x_2 + 3x_4 = 9$ $-2x_1 + 4x_2 + 2x_3 - x_4 = -11$ $4x_1 - 8x_2 + 6x_3 + 7x_4 = -5$	CO1
	(c) State Cayley Hamilton Theorem and use it to simplify and calculate $-A^3 + 4A^2 + 5A - 2I$, where $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ and I is 3*3 Identity matrix.	CO1

Q3	**(Attempt any Two Parts) UNIT-2 (CO2)**	**(5,5)**
	(a) If $z = \sin^{-1}\left(\frac{x^{1/3}+y^{1/3}}{x^{1/2}+y^{1/2}}\right)$, show that $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x \partial y} + y^2\frac{\partial^2 z}{\partial y^2} = \frac{1}{36}(7 + tan^2 z)tanz$	CO2
	(b) Expand $f(x) = \log \sec x$ as far as the terms in x^6.	CO2
a^2	(c) By changing the independent variables x & y to u & v by means of the relation $u = x - ay, v = x + ay$ show that $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2}$ transforms into $4a^2\frac{\partial^2 z}{\partial u \partial v}$.	CO2